

Lec 5

Recap:

- Example $\rightarrow l^p$ (Banach space)
- Non Banach space $(C[0,1], \|\cdot\|_\infty)$
- $(C[a,b], \|\cdot\|_\infty) \rightarrow BS$ $(P[a,b], \|\cdot\|_p)$
- $(B([a,b]), \|\cdot\|_\infty) \rightarrow BS$

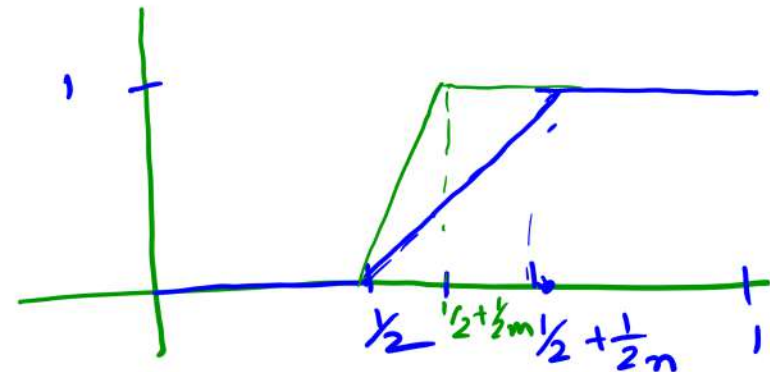
Qn. $(P[a,b], \|\cdot\|_p)$ is a B.S. ? Not. $1 \leq p < \infty$

Today

- $(C[a,b], \|\cdot\|_1)$ is not a B.S.

$$\|x\|_1 = \int_a^b |x(t)| dt \quad |x_n(t)| \leq 1$$

$[0,1]$



For each $n \in \mathbb{N}$.

$$x_n(t) = 0, \quad 0 \leq t \leq \frac{1}{2}$$

$$= \cancel{a}t + \cancel{b}^n, \quad \frac{1}{2} \leq t \leq \frac{1}{2} + \frac{1}{2n}$$

$$= 1, \quad \frac{1}{2} + \frac{1}{2n} \leq t \leq 1.$$

$$m > n \quad \|x_m - x_n\|_1.$$

$$\int_0^1 |x_m(t) - x_n(t)| dt.$$

$$\begin{aligned} &= \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{2n}} |x_m(t) - x_n(t)| dt \leq \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{2n}} |x_m(t)| + |x_n(t)| dt \\ &\leq \underline{2} \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{2n}} dt \leq 2 \cdot \frac{1}{2n} \rightarrow 0 \\ &\quad \text{as } n \rightarrow \infty. \end{aligned}$$

Thus $\|x_m - x_n\|_1 \rightarrow 0$ as $m, n \rightarrow \infty$.

ie (x_n) is a C.S in $C[0,1]$.

Let $\|x_n - x\|_1 \rightarrow 0$ as $n \rightarrow \infty$

$$\int_0^{1/2} |x(t)| dt.$$

$$= \int_0^{1/2} |x_n(t) - x(t)| dt$$

$$\leq \|x_n - x\|_1 \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\int_0^{1/2} |x(t)| dt = 0$$

$$x(t) = 0, \quad 0 \leq t \leq \frac{1}{2}$$

$$\int_{\frac{1}{2} + \frac{1}{2n}}^1 |1 - x(t)| dt$$

$$= \int_{\frac{1}{2} + \frac{1}{2n}}^1 |x_n(t) - x(t)| dt$$

$$\leq \|x_n - x\|_1$$

$$\int_{\frac{1}{2}}^1 |1 - x(t)| dt = 0$$

$$x(t) = 1, \quad \frac{1}{2} \leq t \leq 1$$

$$x(t) \notin C[0, 1]$$

• Ex. $(C[a, b], \|\cdot\|_p)$, $1 \leq p < \infty$ is

NOT a BS.

In fact it is a dense subspace of $L^p[a, b]$.

$$\int_a^b |f(x)|^p dx < \infty.$$

Q Does for an nLS which are not Banach space

~~is~~ $\exists$ a B.S s.t. $nLS \subseteq BS$.

Yes!!

$$X \rightarrow \text{nls. } \|\cdot\|_X$$

Let (x_n) be a C.S. in X .

• $(\|x_n\|)$ is a C.S. in $\mathbb{K}$.
 So $\lim_{n \rightarrow \infty} \|x_n\|$ exists.

$(x_n), (y_n) \in X$ both C.Seq.

• $(\|x_n - y_n\|)$ is C.S. in $\mathbb{K}$.
 ie $\lim_{n \rightarrow \infty} \|x_n - y_n\|$ exists.

$\mathcal{X}$: collection of all Cauchy seq
 in X .

' $\sim$ ' $x = (x_n) \in \mathcal{X}, y = (y_n) \in \mathcal{X}$

$$x \sim y \quad \text{if} \quad \lim_{n \rightarrow \infty} \|x_n - y_n\| = 0$$

' $\sim$ ' is an equivalence reln.

$\tilde{X}$ be the equivalence classes
for an ~~to~~ $(x_n) \in X \rightarrow [(x_n)] \in \tilde{X}$

For $[(x_n)], [(y_n)] \in \tilde{X}$, $\alpha \in \mathbb{K}$

$$[(x_n)] + [(y_n)] = [(x_n + y_n)]$$

$$\alpha \cdot [(x_n)] = [(\alpha x_n)]$$

$(+, \cdot)$ are well defined.

$\tilde{X}(\mathbb{K}) \rightarrow$ linear space.

$\|\cdot\|_*$ defined by $[(0_n)]$

$$\|[(x_n)]\|_* = \lim_{n \rightarrow \infty} \|x_n\|_X \quad \begin{matrix} Q_n = 0 \\ \forall n. \end{matrix}$$

$\|\cdot\|_*$ is a norm (verify).

Then $(\tilde{X}, \|\cdot\|_*)$ is a nls.

Theo. Let X be nls and $\tilde{X}$ be a nls as above.

Then $\tilde{X}$ is a completion of X .

Completion: $X \rightarrow \text{nls.}$

$Y \rightarrow \text{BS.}$
 Y is a completion of X if $\exists$

isometry,
 $T: X \rightarrow Y$ s.t. $R(T)$ is dense
 $\nwarrow$ in Y .
 (range of T)

Claim
 $\tilde{X}$ is a B.S.

Let $(\tilde{x}_k)$ be a C.S. where

$\tilde{x}_k \in \tilde{X}$ that is,

$\tilde{x}_k = [(x_{k,n})]$, $(x_{k,n})$ is
 a C.S. in X .

$\rightarrow \tilde{x}_1 = [(x_{1,n})]$, $(\underline{x_{1,n}})$ is a C.S.
 in X

$\tilde{x}_2 = [(x_{2,n})]$, $(x_{2,n})$ is a C.S.
 in X .

$$\epsilon = 1 \quad \exists N_1 \quad \text{s.t.}$$

$$\|x_{1,N_1} - x_{1,p}\| < 1, \quad \forall p \geq N_1$$

$$\epsilon = \frac{1}{2} \quad \exists N_2 \quad \text{s.t.}$$

$$\|x_{2,N_2} - x_{2,p}\| < \frac{1}{2}, \quad \forall p \geq N_2$$

Then continuing we have

$$\|x_{k,\underline{N_k}} - x_{k,p}\| < \frac{1}{k}, \quad \forall p \geq N_k.$$

— ①

Now consider

$$\|x_r, N_r - x_m, N_m\|$$

$$= \|x_{r, N_r} - \underline{x_{r, n}} + \underline{x_{r, n}} - \underline{x_{m, n}} + \underline{x_{m, n}} - x_{m, N_m}\|$$

$$\leq \underbrace{\|x_{r, N_r} - x_{r, n}\|}_{\text{red underline}} + \|x_{r, n} - x_{m, n}\| + \underbrace{\|x_{m, n} - x_{m, N_m}\|}_{\text{red underline}}$$

$$\leq \frac{1}{r} + \frac{1}{m} + \underbrace{\|x_{r, n} - x_{m, n}\|}_{\text{blue oval}} \quad \forall n \geq \max\{N_r, N_m\}$$

$$< \frac{1}{r} + \frac{1}{m} + \epsilon \quad \begin{matrix} n > m & m < n \\ m > N_\epsilon & n < N_\epsilon \end{matrix} \quad \textcircled{2}$$

$(\tilde{x}_k)$ is a $\mathcal{C}\mathcal{S}$ in $\tilde{\mathcal{X}}$

$$\forall \epsilon > 0$$

$$\|\tilde{x}_n - \tilde{x}_m\|_\alpha < \epsilon \quad \begin{matrix} n > m \\ m > N_\epsilon \end{matrix}$$

$$\therefore \lim_{n \rightarrow \infty} \|x_{r,n} - x_{m,n}\| < \epsilon$$

$$\forall r > m \\ m > N_\epsilon.$$

$$\|x_{r,n} - x_{m,n}\| < \epsilon \text{ for arbitrary } n. \quad \text{--- (3)}$$

So from (2) & (3) we get

$(x_{n, N_n})_{n \in \mathbb{N}}$ is a C.S in X .
--- (4)

Let $a_n = x_{n, N_n}$

(a_n) is a C.S in X .

Then

$$[(a_n)] \in \tilde{\mathcal{X}}$$

$$\tilde{a} = [(a_n)] \in \tilde{\mathcal{X}}$$

Claim

$$\tilde{x}_k \xrightarrow{\|\cdot\|_k} \tilde{a} \quad \text{as } k \rightarrow \infty$$

$$\text{i.e., } \|\tilde{x}_k - \tilde{a}\|_k \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

$$\|\tilde{x}_k - \tilde{a}\|_k = \lim_{n \rightarrow \infty} \|x_{k,n} - a_n\|_x$$

$$= \lim_{n \rightarrow \infty} \|x_{k,n} - x_{n,N_n}\|_x$$

$$\underline{k > n}$$

$$\|x_{k,n} - x_{n,N_n}\|_X$$

$$= \|x_{k,n} - x_{k,N_k} + x_{k,N_k} - x_{n,N_n}\|$$

$$\leq \|x_{k,n} - x_{k,N_k}\| + \|x_{k,N_k} - x_{n,N_n}\|$$

$$\leq \frac{1}{k} + \|x_{k,N_k} - x_{n,N_n}\| \quad \forall n > N_k$$

Then taking $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \|x_{k,n} - x_{n,N_n}\|_X$$

$$\underline{k > n}$$

$$\leq \frac{1}{k} + \epsilon$$

$$\|\tilde{x}_k - \tilde{a}\|_* \leq \frac{1}{k} + \epsilon$$

$$\leq \epsilon' \quad \text{for all}$$

sufficiently
large k .

$$\lim_{k \rightarrow \infty} \tilde{x}_k = \tilde{a} \quad \text{in } \|\cdot\|_*.$$

$\therefore \tilde{X}$ is a B.S.

• $\tilde{X}$ is a completion.

$$T: X \rightarrow \tilde{X}$$

$$x \in X$$

$$\tilde{y}_x = [(x)] \text{ — equivalence}$$

class of const seq

$$(x, x, \dots)$$

$$Tx = \tilde{y}_x$$

$$\|Tx\|_* = \|\tilde{y}_x\|_*$$

$$= \lim_{n \rightarrow \infty} \|x\|$$

$$= \|x\|$$

T is an isometry.

• $R(T)$ is dense in $\tilde{X}$.

$$\text{Let } \tilde{x} = [\underline{(x_n)}] \in \tilde{\mathcal{X}}$$

$$\text{Let } \epsilon > 0 \quad \exists N \in \mathbb{N} \text{ s.t.}$$

$$\|x_m - x_n\| < \epsilon \quad \forall n \geq N.$$

$$u_n = x_N \quad \forall n.$$

$$\therefore \tilde{u} = [(u_n)] \in \mathcal{R}(T)$$

$$\underline{\|\tilde{x} - \tilde{u}\|_*} = \lim_{n \rightarrow \infty} \|x_n - u_n\|$$

$$= \lim_{n \rightarrow \infty} \|x_n - x_N\|$$

$$< \epsilon$$

$\therefore R(T)$ is dense in $\tilde{X}$.

Hence $\tilde{X}$ is a completion of X .

Ex. $(C_0, \|\cdot\|_p)$ $1 \leq p < \infty$.

completion $(l^p, \|\cdot\|_p)$

$(P[a, b], \|\cdot\|_\infty)$

$(C[a, b], \|\cdot\|_\infty)$

$C^k[a, b]$

Sobolev
norm.

Recall.

$$X \rightarrow \text{v. sp.}$$

$$M \subseteq X$$

$$\underline{\underline{0 \in M.}}$$

subspace.

Define

$$x, y \in X \quad ' \sim '$$

$$x \sim y, \quad x \rightarrow y \in M.$$

$$' \sim ' \quad \text{eq} \quad \text{rel}_M.$$

$$[x] = \{y \in X \mid x - y \in M\}$$

$$= x + M \text{ cosets}$$

$$X/M = \{x + M \mid x \in X\}$$

$$= \{[x] \mid x \in X\}$$

Define .

$$(x + M) + (y + M)$$

$$= x + y + M.$$

$$\alpha(x + M) = \alpha x + M.$$

$$\alpha[x] = [\alpha x]$$

Check these are well defined !!

X/M is a linear space.

Define

$$\|x + M\|_*$$

$$= \text{dist}(x, M)$$

$$= \inf \{ \|x - m\|_x \mid m \in M \}$$

$$\left(\begin{array}{l} \text{dist}(x, M) = 0 \\ \Rightarrow x \in \overline{M} \end{array} \right)$$

$\| \cdot \|_*$ is a seminorm

but if M is a
closed subspace then

$\|\cdot\|_*$ is a norm.

- If M is a closed
subspace of X then
 $\dagger$ X/M is a nls.
 $\|\cdot\|_*$.